

Chest X-Ray Pneumonia Classification

Cynthia Du, Angelina Pace

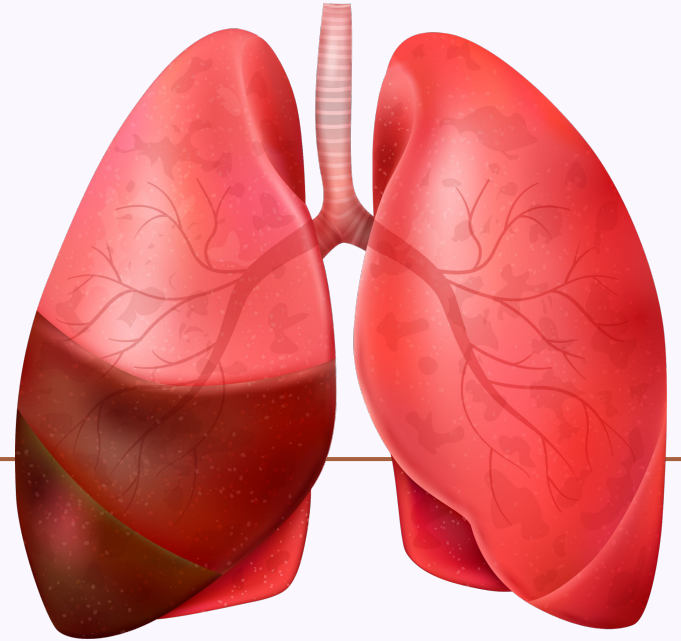




Table of contents

01

Research Question

What is the problem/research question, and why is it important? Why is it interesting to tackle the problem with a neural network model?

02

Background

What has been done about it & what gap/aspect are you addressing?

03

Our Data

What is your dataset? Is it appropriate to test your hypothesis/research question?



04

Our Model

What is our model? Why did we choose it for this problem? What is our input/output? How are they defined?

05

Our Procedures

What is the architecture? What is the procedure of train/validate/testing in our model?

06

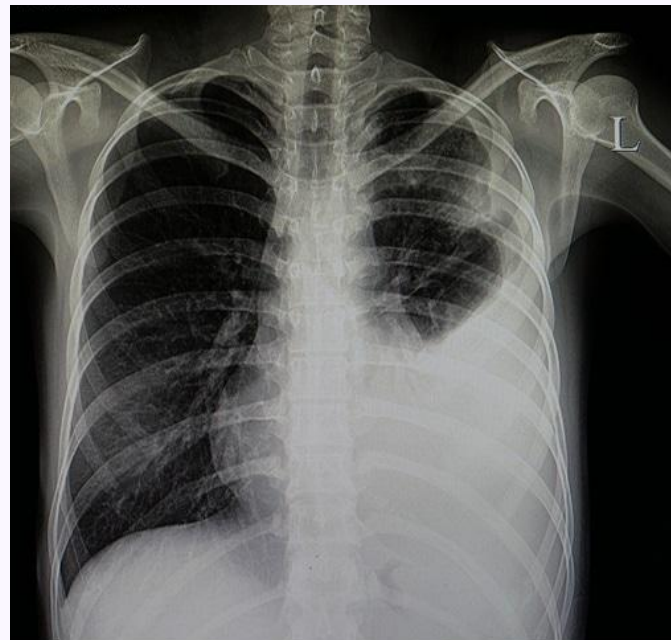
Performance Evaluation

How do you evaluate performance? When does the model fail/succeed and why?

Background

Pneumonia Research

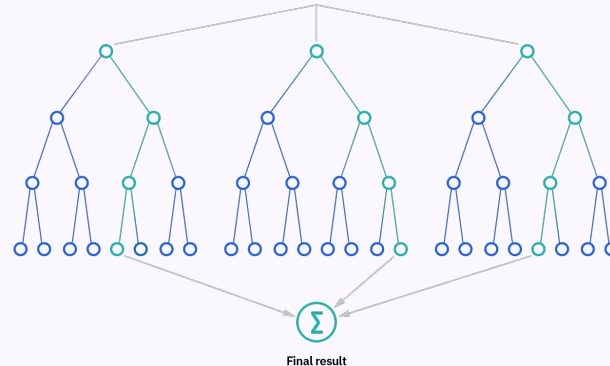
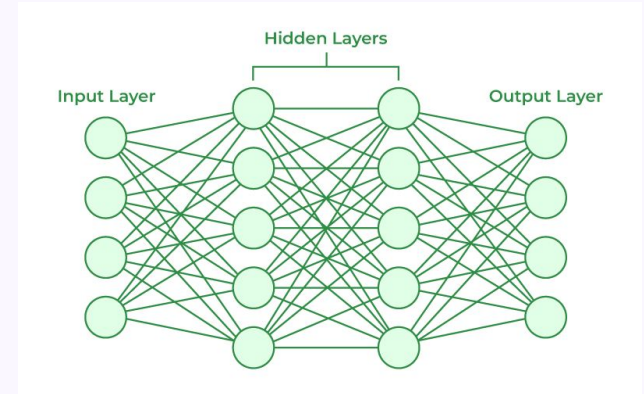
- Pneumonia: infection that causes inflammation in the air sacs in one or both lungs (fill with fluid or pus)
- Diagnosis: physical exam, chest X-rays, blood tests, pulse oximetry
- Studies released by PubMed Central (PMC), and NIH use AI tools to analyze chest x-rays (95% sensitivity, 90% specificity in identifying pneumonia)

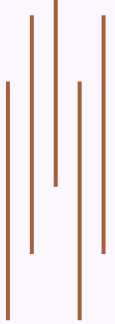


Background

ML Models in the Medical Field

- Neural networks, CNNs, and decision trees already in use to analyze healthcare data to improve diagnostics, treatment, and patient outcomes.
- Key applications: detecting tumors in MRIs, CT scans, predicting mortality rates, optimize treatment sequences, etc.





Background

Current Challenges

In Healthcare:

- Data privacy & bias
- Implementation
- Explainability

In Public Domain:

- Medical imaging datasets are often limited/imbalanced

Our Goal

- Implement & analyze a machine learning model
 - Experiment with different hyperparameters & training techniques
 - Evaluate model performance beyond accuracy
- 

Research Question

Our Question

Can we develop an automated system that can assist with pneumonia detection more efficiently and accurately?

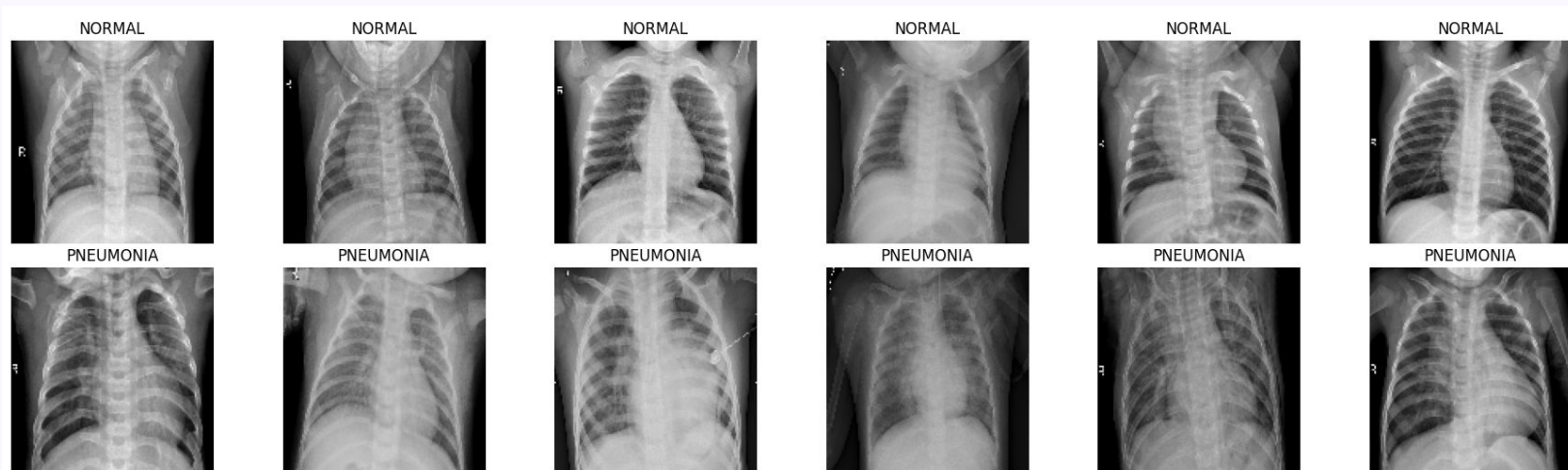
- Pneumonia is a serious respiratory infection that can be life-threatening if not detected and treated early
- Diagnosis relies on radiologists interpreting chest X-rays → time consuming, subject to human error

Why?

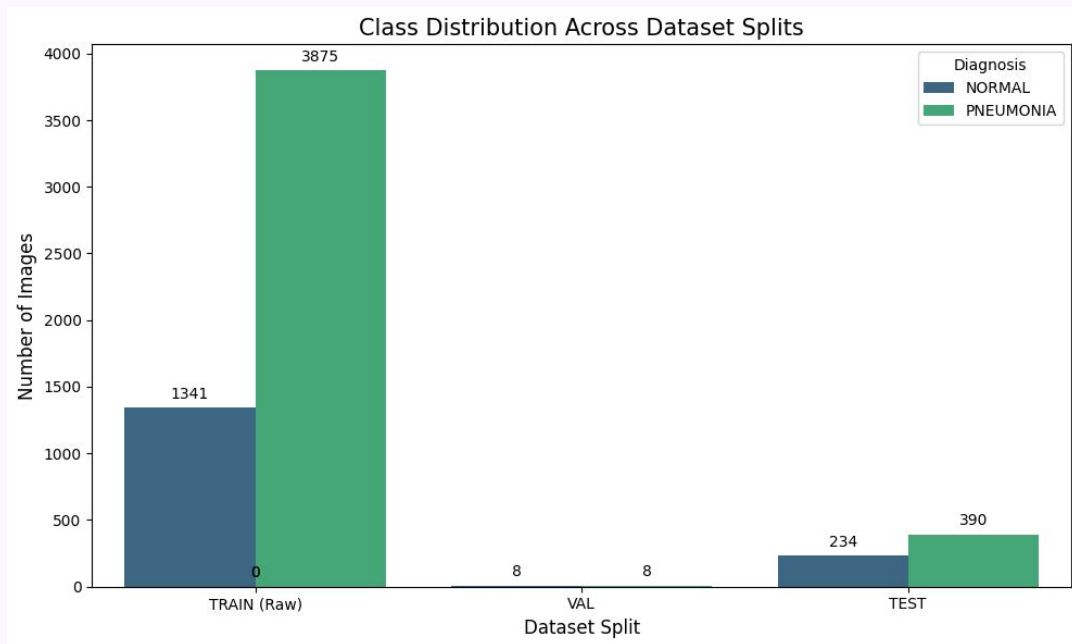
- Medical images contain complex spatial patterns
 - CNNs designed to analyze image data by automatically learning hierarchical features (edges, textures, higher-level visual patterns) ~ (lung opacity, fluid accumulation, abnormal tissue density)
- Allows the model to learn these diagnostic features directly from the raw data > predefined rules
 - Can be applied to real-world healthcare

The Dataset

- **5,863 chest X-ray images** that were taken from pediatric patients aged one to five at the Guangzhou Women and Children's Medical Center in Guangzhou, China
- Pre-organized into training, testing, and validation datasets, with the training and validation data labeled as either **"PNEUMONIA"** or **"NORMAL"**
- Images are 128 x 128



The Dataset



Training

- 3875 Pneumonia Images
- 1341 Normal Images

Testing

- 390 Pneumonia Images
- 234 Normal Images

Validation

- 8 Pneumonia Images
- 8 Normal Images

74% of images classified as Pneumonia

Fixing Data Imbalance

A big issue with our dataset is that the validation set is very small and the training set is highly imbalanced. This could cause our model to predict 'pneumonia' just because there are more instances

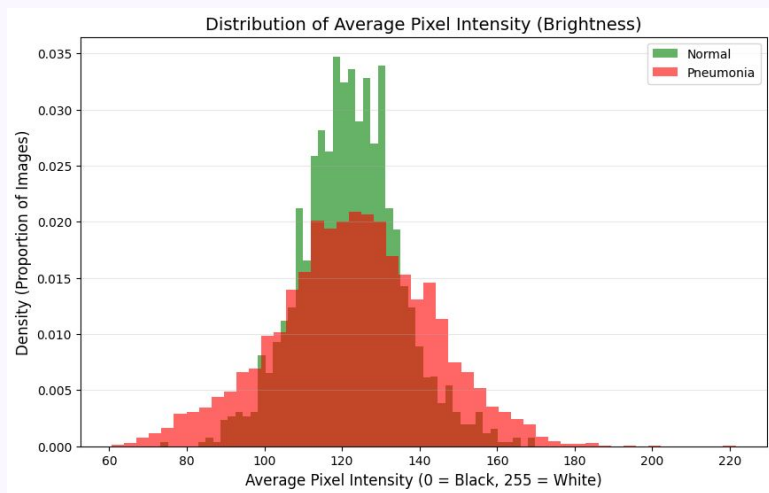


We fixed the imbalance so the training set is 50% pneumonia and 50% normal

Ensured that 15% of the data is reserved for a statistically significant validation set that mirrors the true distribution of the classes

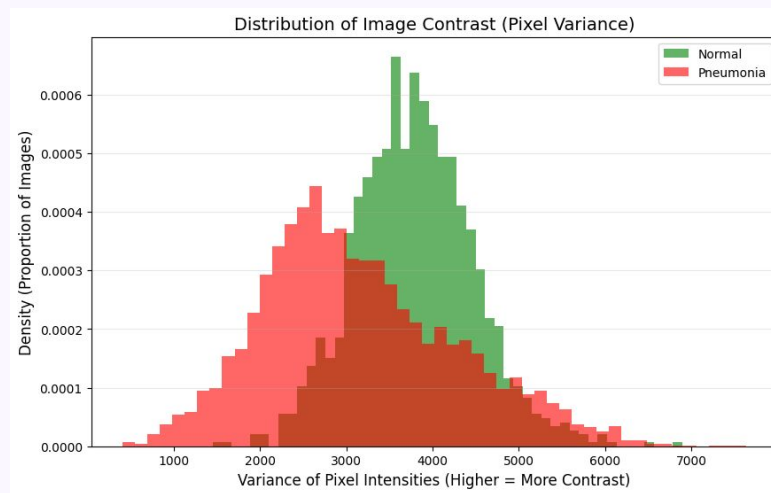
Statistical Differences Between Classes

Pixel Intensity (Brightness)



Pneumonia images show greater variability in average brightness, whereas normal images are much more consistent

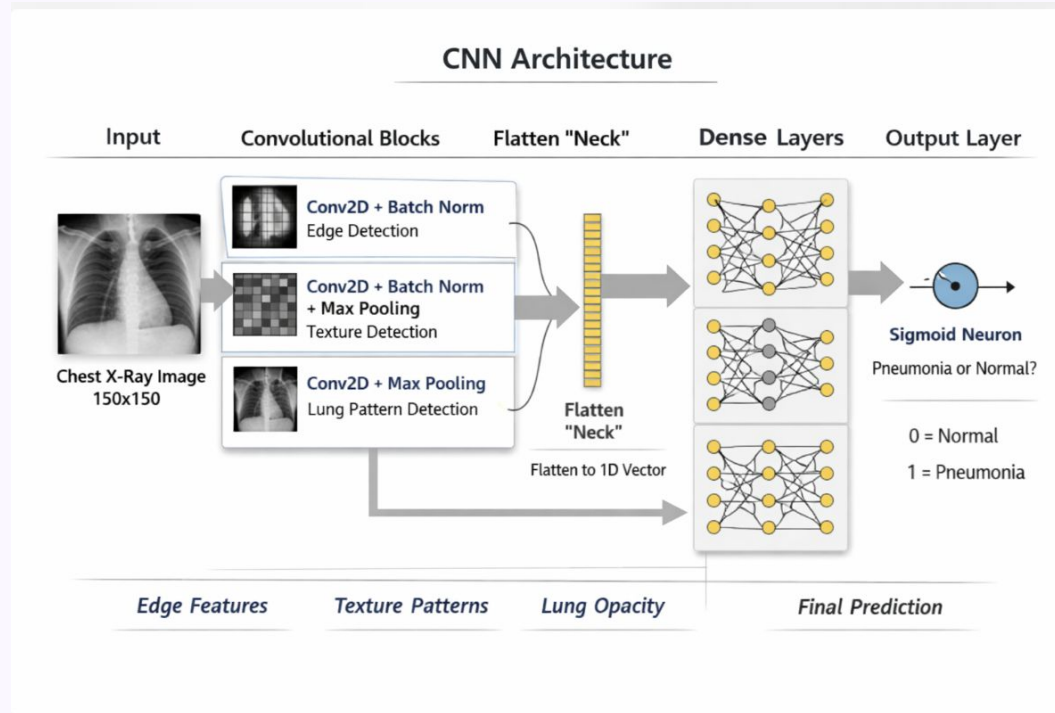
Image Contrast (Variance)



Pneumonia's average contrast is lower than those of normal X-ray images - images are generally less sharp, making the image appear more hazy.

Modeling

CNN Model



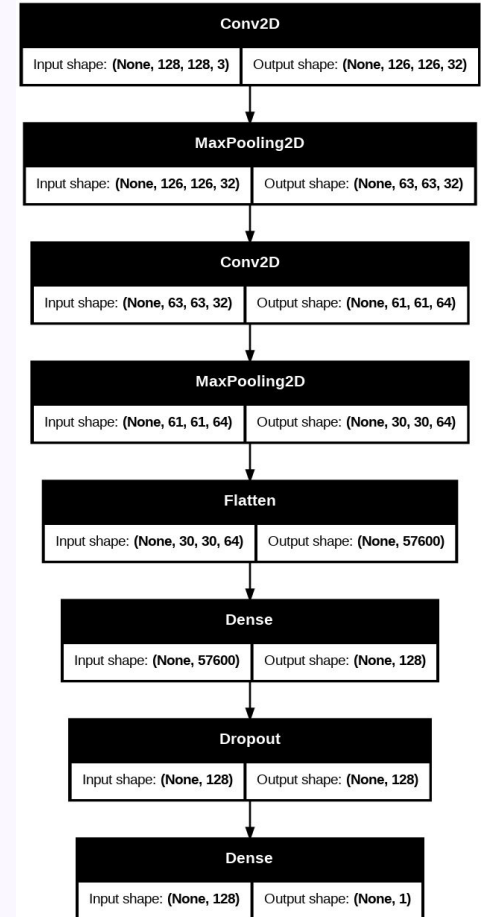
CNN Model

Feature Extraction

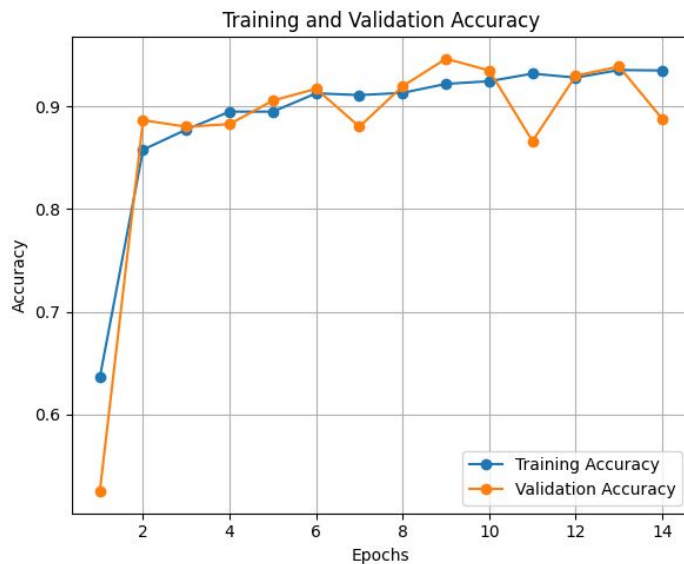
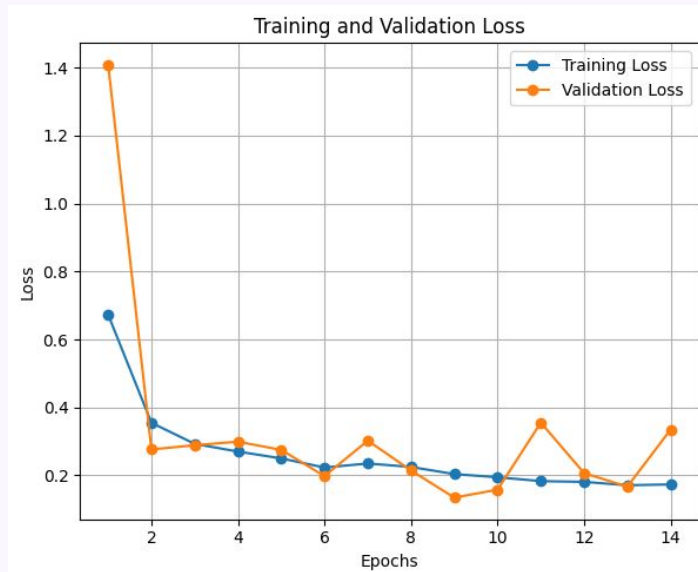
- **Sequential Convolutional Layers:** 2 primary convolution blocks (32 and 64 filters) to detect everything from sharp rib edges to the faint, “cloudy” opacities indicative of pneumonia (focuses on lung textures)
- **Batch Normalization:** Each convolutional layers is followed by batch normalization to stabilize the learning process and mitigate “internal covariate shift” → higher learning rates, more robust training
- **Max Pooling:** Downsampling layers used to reduce spatial dimensions of the feature maps → model focuses on most significant diagnostic markers instead of background noise

Classification

- **Regularized Dense Network:** Fully connected layer with 128 neurons = primary decision-maker. 50% dropout rate = learn redundant representations and prevent overfitting on training set
- **Sigmoid Output:** Provides probability score (0 to 1) for binary classification which is then mapped to “Normal” or “Pneumonia”



Results



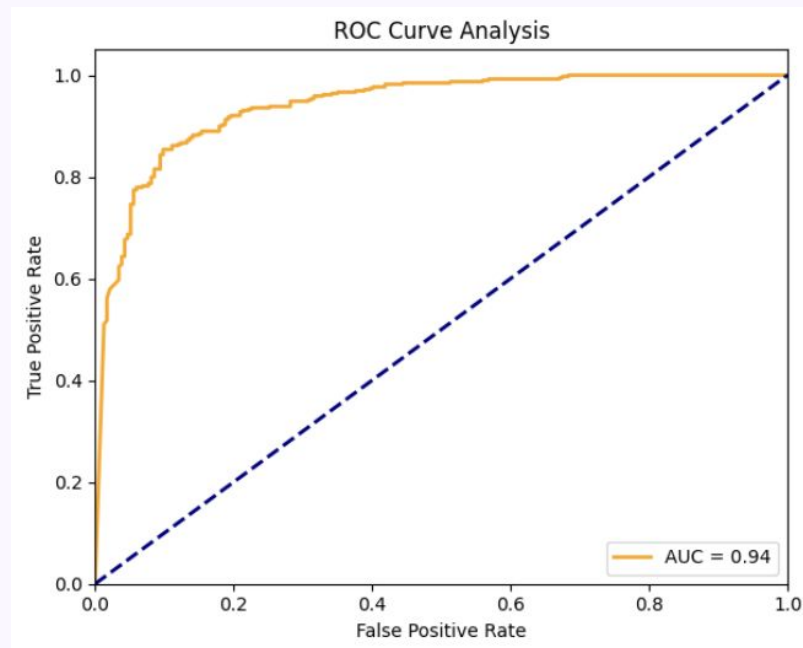
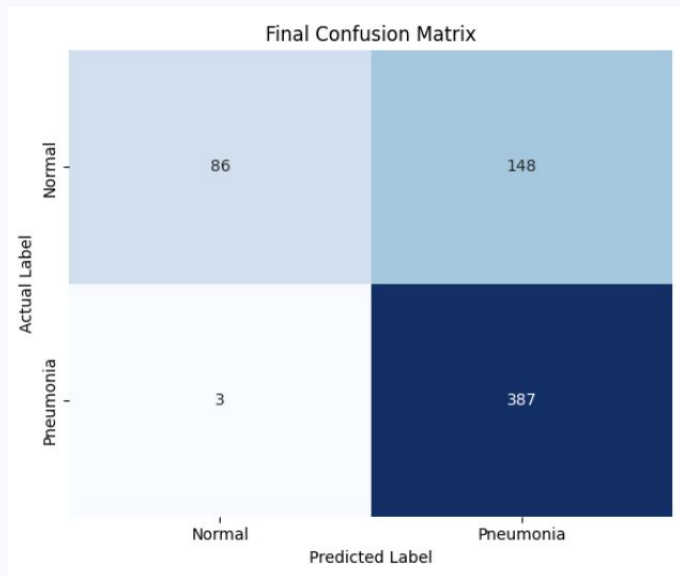
Model performing well, no evidence of overfitting

Results

Test Accuracy: 0.7580

Test Precision: 0.7234

Test Recall: 0.9923



Results

20/20 ————— 4s 214ms/step

Predicted: Pneumonia
Actual: Normal



Predicted: Pneumonia
Actual: Normal



Predicted: Normal
Actual: Normal



Predicted: Pneumonia
Actual: Normal



Predicted: Normal
Actual: Normal



Predicted: Pneumonia
Actual: Normal



Predicted: Pneumonia
Actual: Normal



Predicted: Normal
Actual: Normal



Predicted: Pneumonia
Actual: Normal



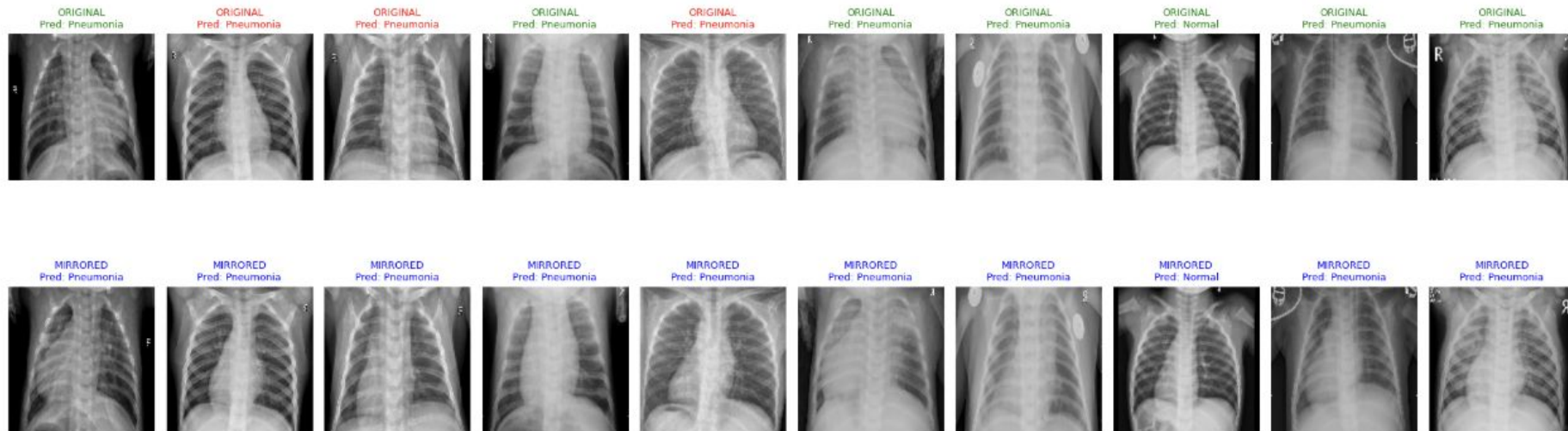
Predicted: Pneumonia
Actual: Normal



Mirrored Image Test

To ensure clinical reliability:

- Evaluate performance on horizontally flipped images to verify that the model's diagnosis remains consistent regardless of the patient's orientation on the side of the chest displaying infection markers

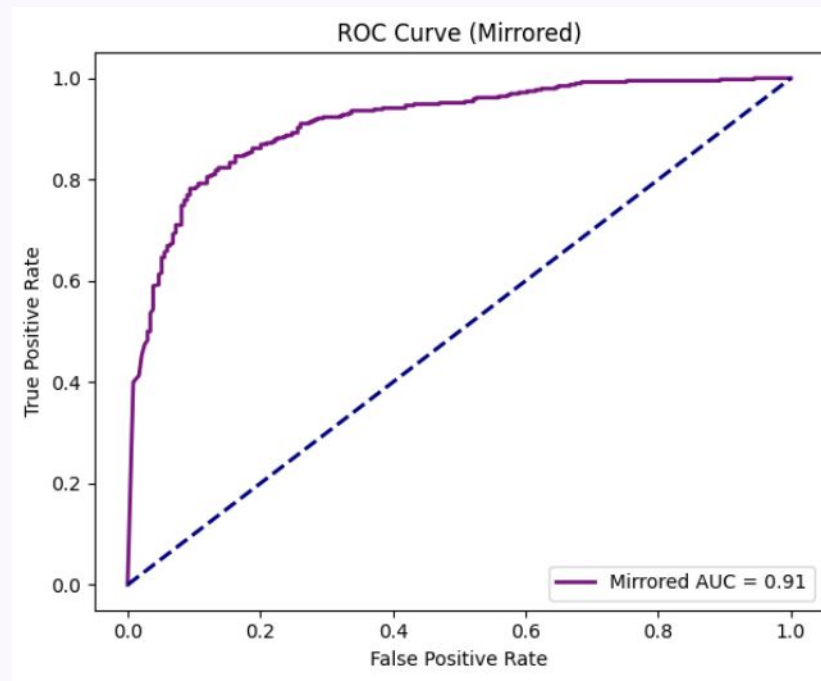
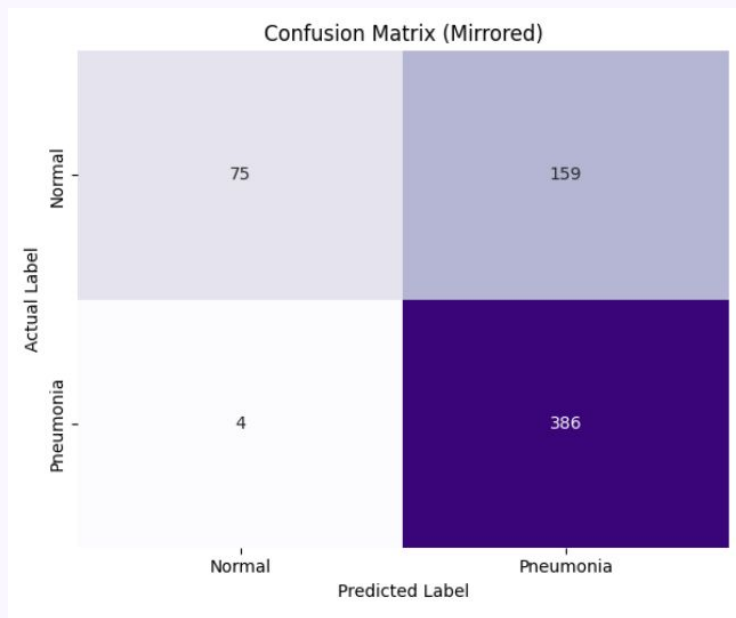


Mirrored Image Test Results

Test Accuracy: 0.7388

Test Precision: 0.7083

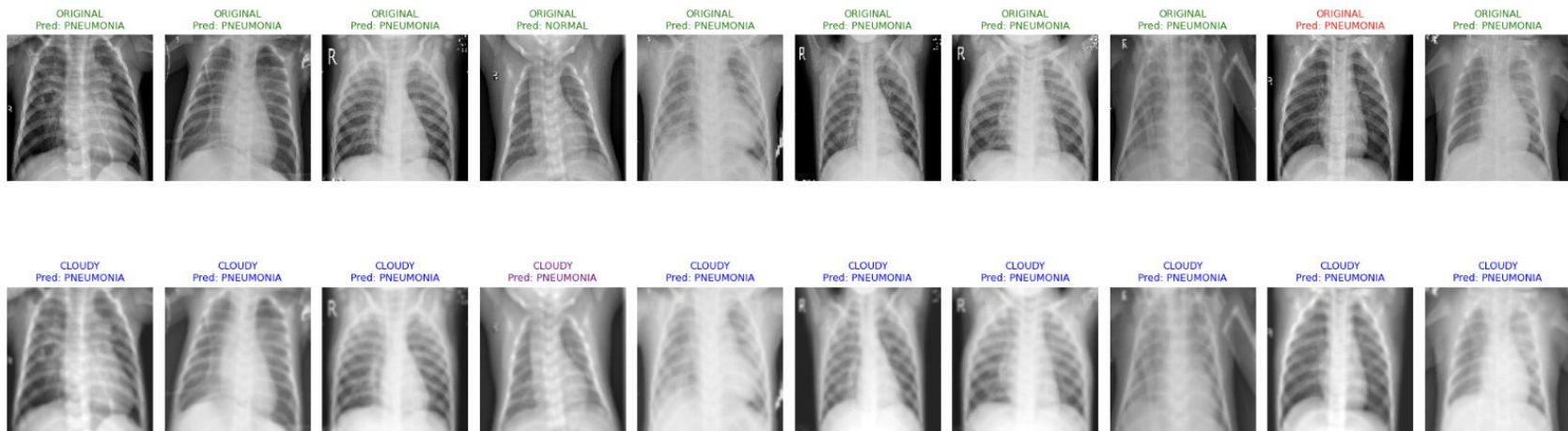
Test Recall: 0.9897



“Cloudy” Image Test

To ensure clinical reliability:

- Use Synthetic Data Augmentation (apply Gaussian Blur) to overlay a semi-transparent white “mask” to mimic white, hazy “clouds” (infiltrates) on an X-ray (what physicians look at when determining diagnosis)

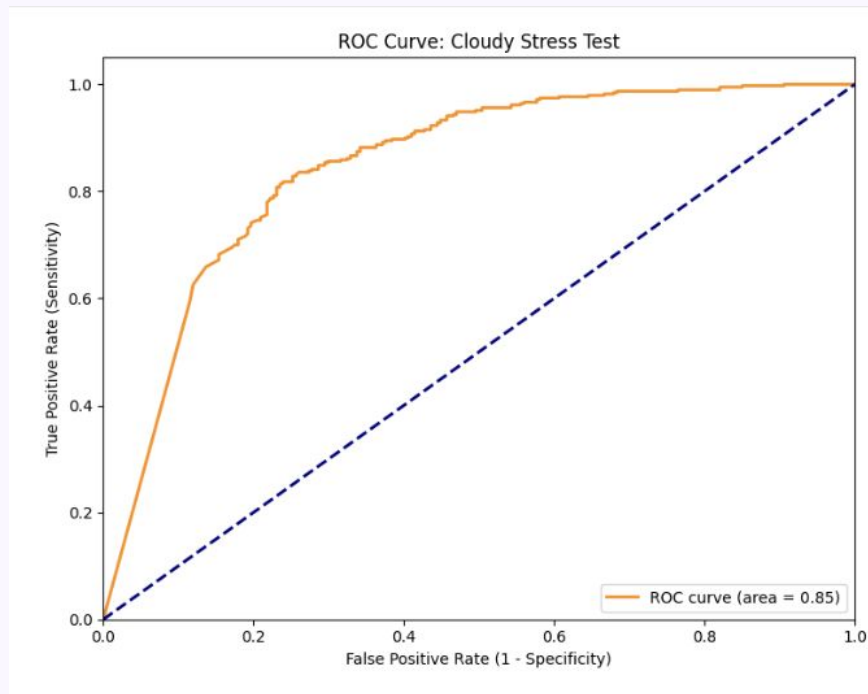
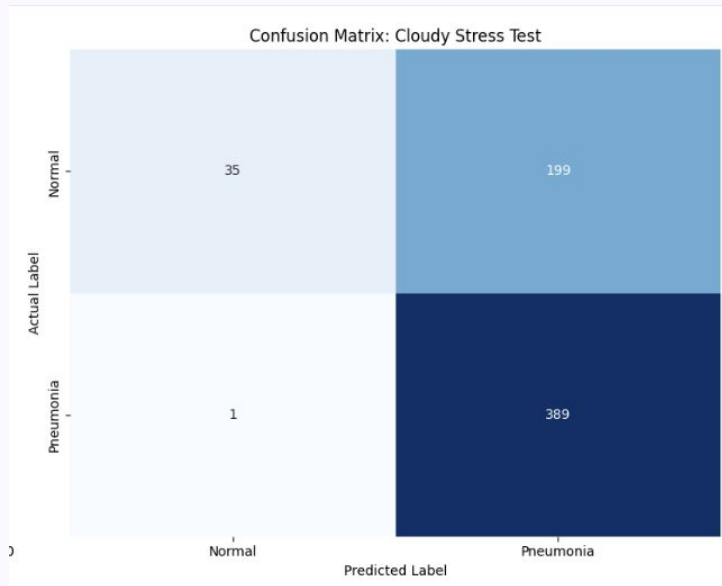


“Cloudy” Image Test Results

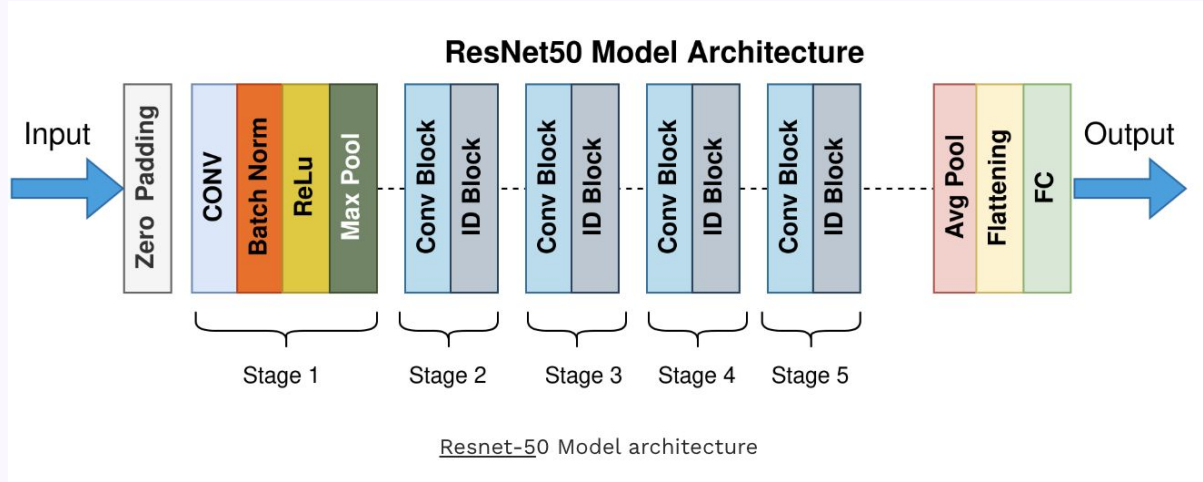
Test Accuracy: 0.6795

Test Precision: 0.6616

Test Recall: 0.9974



Resnet-50 Model



ResNet (Residual Network) is a deep convolutional neural network introduced by He et al. in 2015, designed to make training very deep models easier and more effective

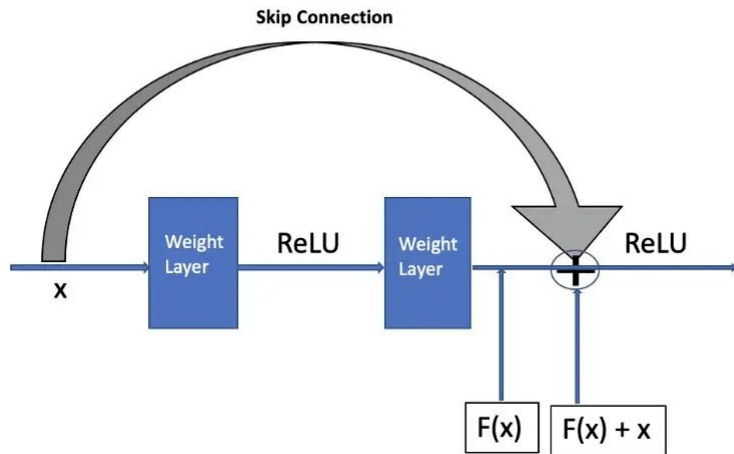
- Resnet-50: 49 convolutional layers and 1 fully connected layer

Resnet-50 Architecture

The architecture of ResNet50 is divided into four main parts:

- **The convolutional layers:** extracts features from the input image using kernels
 - Fundamental building block of a CNN
 - Detects patterns
- **The identity block:** process and transform extracted features
 - Allows data to bypass layers through a "shortcut" or "skip connection"
 - Keeps the input and output dimensions the same, using a formula of $y = F(x) + x$, where x is the input and $F(x)$ is the residual mapping
- **The convolutional block:** process and transform extracted features
 - Designed for computer vision problems like this
 - Combines convolutional layers with activation functions and pooling layers
- **The fully connected layers:** learn non-linear combinations of features extracted by convolutional layers to make the final classification
 - Every neuron connects to every other neuron

Vanishing Gradient Problem



Resnet-50 solves the vanishing gradient problem (gradients become extremely small, which makes it difficult for layers to learn and improve)

Skip connections address this problem by allowing the information to flow directly from the input to the output of the network, bypassing one or more layers. This allows the network to learn residual functions that map the input to the desired output, rather than having to learn the entire mapping from scratch.

- In Resnet-50, these connections are used in the identity block and convolutional block

Data Processing

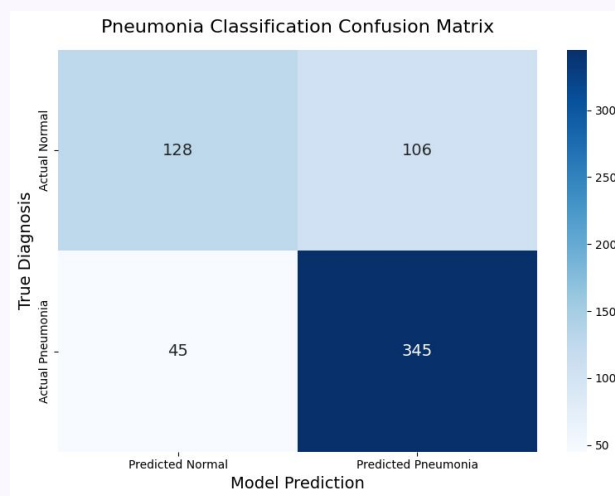
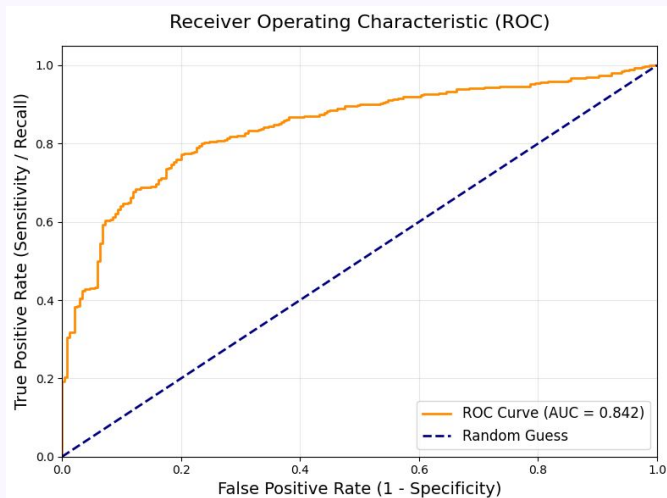
ResNet expects all images to be of shape (224, 224, 3). This means that we had to resize all our images to be of size 224 x 224. Furthermore, since our dataset consists of grayscale X-rays, we had to convert them to RGB format to match ResNet's expected 3-channel input.

- Implemented a dynamic `tf.data.Dataset` pipeline to automatically resize chest X-rays to the 224x224 resolution required by our model
- Applied random rotations, translations, zooms, and contrast adjustments during training to artificially expand dataset variance and prevent overfitting.



Base Model

- Leveraged ResNet50 (pre-trained on ImageNet) as a frozen feature extractor to identify complex edges and textures without massive computational overhead.
- Replaced the original output layers with a specialized Dense network tailored for binary diagnostic classification.
- Included a 50% Dropout layer in the custom head to force distributed learning and prevent reliance on specific neural pathways.



Fine Tuning

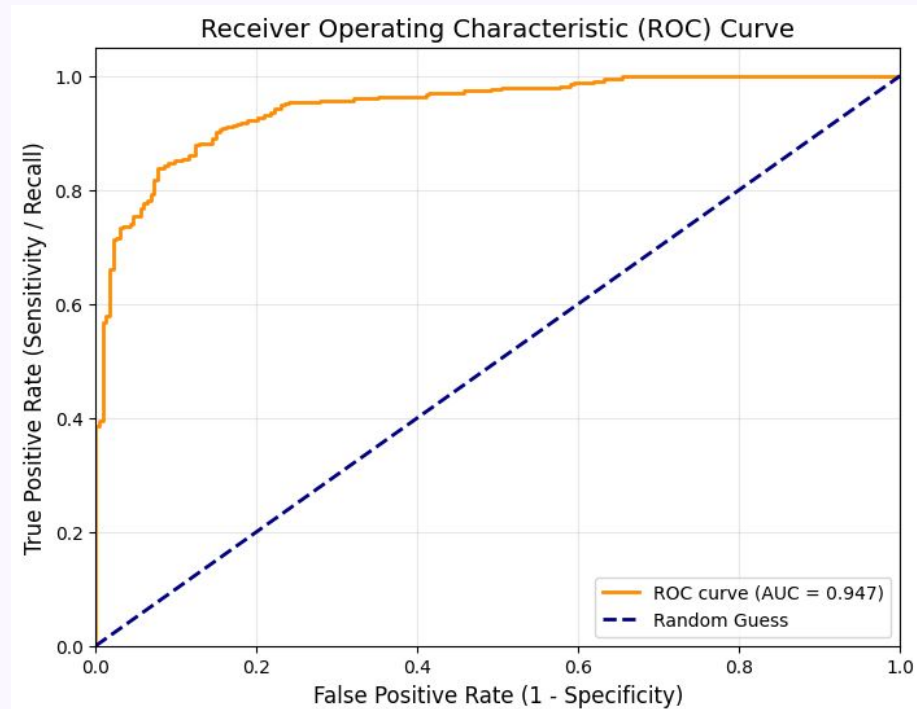
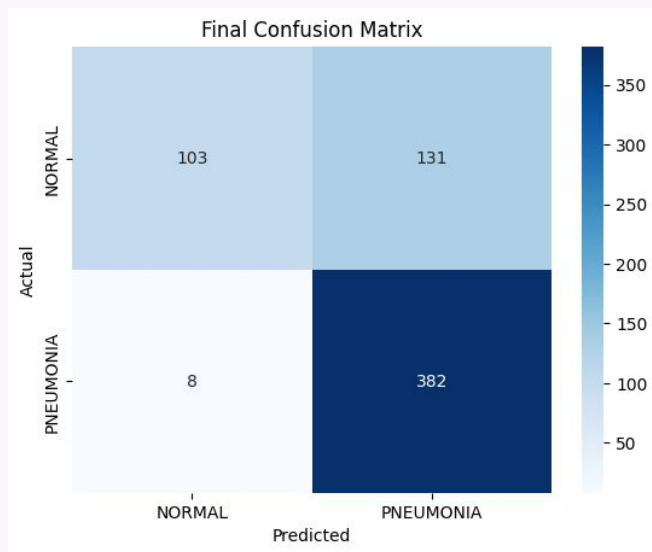
- Unfroze the final 32 layers of the ResNet-50 backbone (Layer 143 onwards) to adapt its abstract feature detection specifically to pneumonia opacities.
 - Allows it to learn the highly specific texture of chest X-rays
 - Early layers still frozen since they detect basic shapes and edges
- Drastically reduced the learning rate 1×10^{-5} to carefully adjust the network without triggering "catastrophic forgetting" (destroying all pre-trained weights) of its foundational ImageNet features.
- Implemented a ReduceLROnPlateau scheduler to automatically halve the learning rate if validation loss stagnated, preventing the model from overshooting the optimal weights.
 - Allows model to take smaller, more careful steps toward the optimal diagnostic accuracy

Results

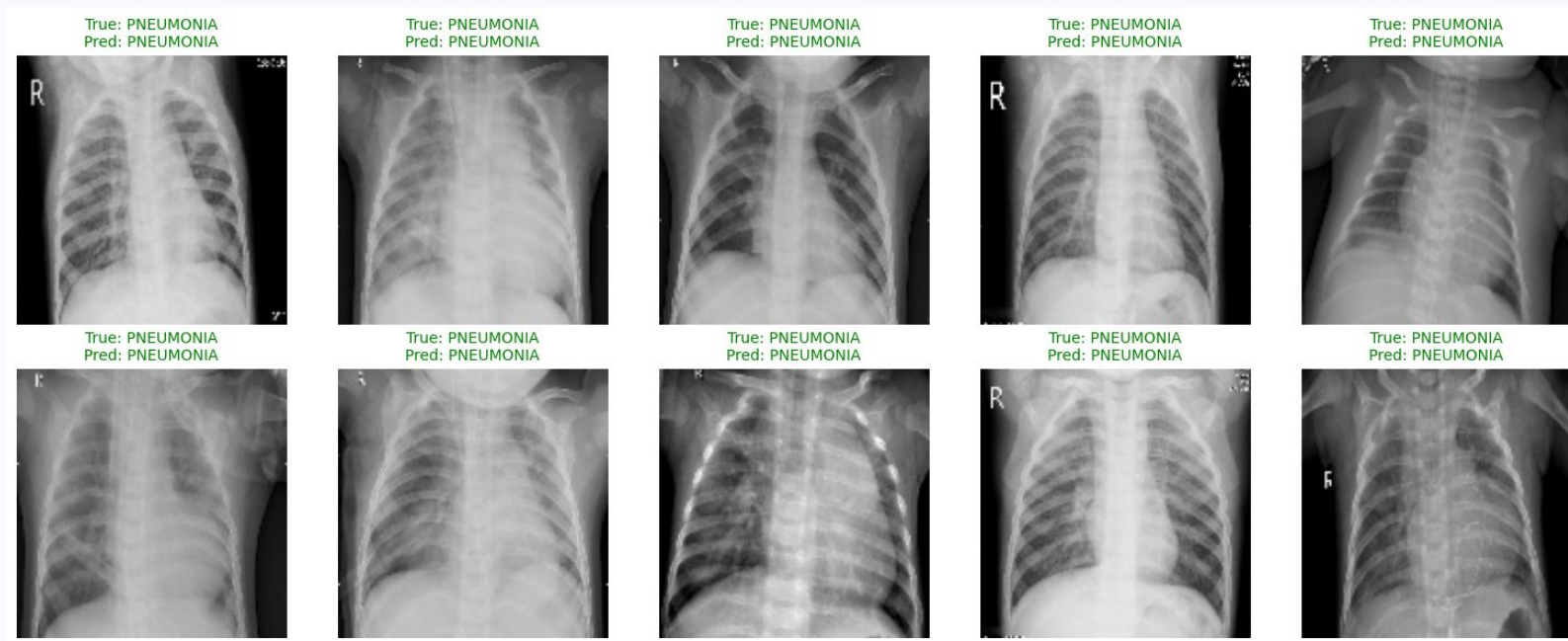
Test Accuracy: 0.7772

Test Precision: 0.7446

Test Recall: 0.9795

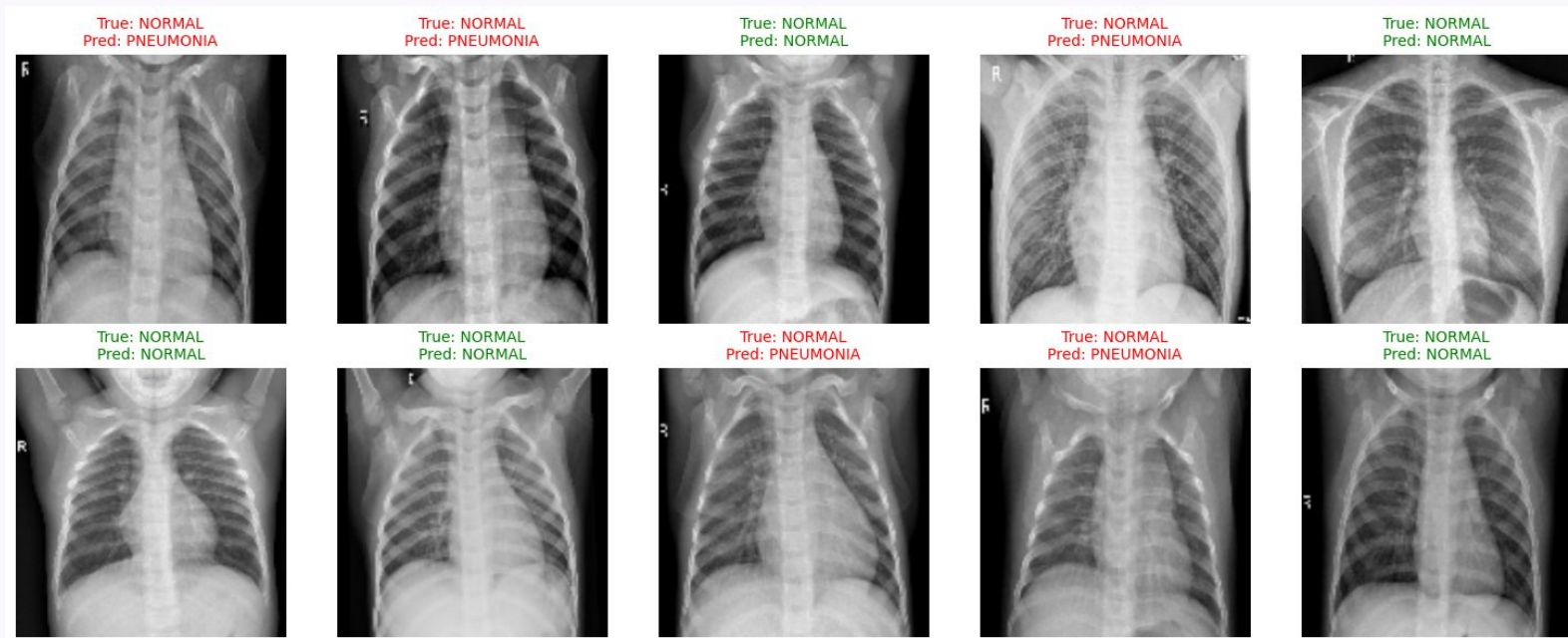


Results



The model casts a very wide net to ensure it catches almost every single case of pneumonia (high recall), but the trade-off is that it occasionally flags healthy lungs as sick (lower precision)

Results



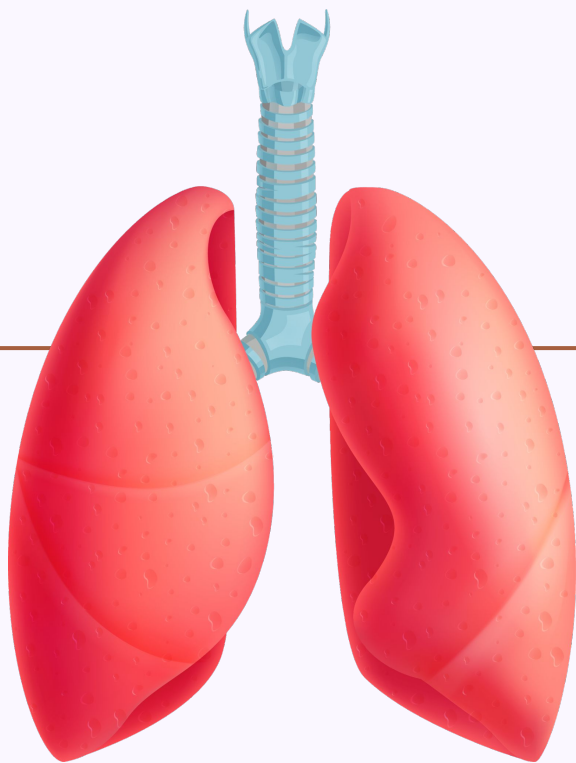
The model casts a very wide net to ensure it catches almost every single case of pneumonia (high recall), but the trade-off is that it occasionally flags healthy lungs as sick (lower precision)

Real life impact

High recall is more important in the context of our problem - nearly 98% means that the model only misses about 2% of sick patients (False Negatives)

In medical screening, sending a sick person home because the model said they were "Normal" is the worst-case scenario, and our model avoids that





Key Findings

- High diagnostic baseline
- Robustness to clinical noise (cloudy test)
- Vulnerability to spatial inversion (mirror test)
- Precision-recall trade-off

Limitations



Biased Data

Small age/region range

- Lung structure & bone density could be different as these vary (environmental baselines)



Class Imbalance

Dataset has more “Pneumonia” cases than “Normal”

- Had to select a subset of the dataset to keep training set balanced



Lack of Explainability

Normal vs. Pneumonia, no “Why?”

- No heat map in our model

Sensitivity to global spatial transformations

- Model treats as a completely new data point



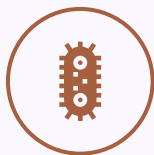
No Clinical Context

Pneumonia diagnosis are not exclusively dependent on chest x-rays

- Patient history, comparison, physical exam

Difference between pattern recognition and clinical reasoning

Future Work



Data Expansion

Acquire larger, more diverse datasets from multiple hospitals to improve model generalization and reduce the need for strict undersampling.



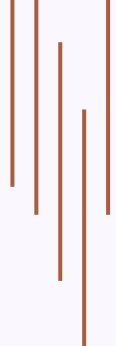
Hyperparameter Tuning

Implement automated search strategies (e.g., Grid Search or Bayesian Optimization) to systematically identify the optimal learning rates, dropout ratios, and fine-tuning depths.



Clinical Context

Integrate interpretability tools (e.g. Grad-CAM) to visually highlight pneumonia opacities, and incorporate patient metadata (age, symptoms) for a more holistic diagnostic tool.



Thank you!

Questions?

